Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-10(P1) | Date: 30-04-26
Time: 3 Hrs | 2021_P1 | Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**D**	2	**D**	3	**A**	4	**D**	5	**0.63**
6	**0.65 TO 0.66**	7	**6**	8	**3**	9	**1.41**	10	**0.66 TO 0.67**
11	**ABCD**	12	**ACD**	13	**ABC**	14	**CD**	15	**ACD**
16	**ABD**	17	**3**	18	**5**	19	**6**		

PHYSICS

20	**D**	21	**C**	22	**A**	23	**D**	24	**326**
25	**30.50 TO 31.50**	26	**0.22 TO 0.23**	27	**0.25 TO 0.27**	28	**1.62 TO 1.63**	29	**1.35 TO 1.37**
30	**ABCD**	31	**ACD**	32	**AC**	33	**C**	34	**ABCD**
35	**ABC**	36	**2**	37	**10**	38	**8**		

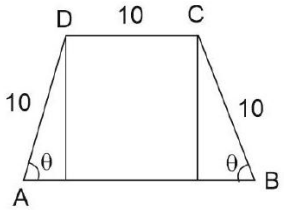
CHEMISTRY

39	**C**	40	**A**	41	**C**	42	**D**	43	**27.93**
44	**514**	45	**9**	46	**14**	47	**51**	48	**1.20**
49	**D**	50	**AB**	51	**ACD**	52	**ABC**	53	**ABC**
54	**ABD**	55	**800**	56	**3**	57	**28**		

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SOLUTIONS
MATHS

1. $A=\frac{1}{2}(10+10+20\cos\theta)10\sin\theta$



$A=100(1+\cos\theta)\sin\theta\ A^2=10000(1+\cos\theta)^3(1-\cos\theta)\Rightarrow A^2=10000t^3(2-t)$, where $t=1+\cos\theta>0$

$\frac{\frac{t}{3}+\frac{t}{3}+\frac{t}{3}+(2-t)}{4}\geq\left(\frac{t^3}{27}(2-t)\right)^{1/4}\Rightarrow t^3(2-t)\leq\frac{27}{16}$ and equality holds if

$\frac{t}{3}=2-t\Rightarrow t=\frac{3}{2}\Rightarrow\theta=60^\circ\Rightarrow A^2\leq\frac{270000}{16}\Rightarrow A\leq 75\sqrt{3}$

2. $\cos x=A(3\cos x+4\sin x)+B(-3\sin x+4\cos x)$

$4A-3B=0\ 3A+4B=1\Rightarrow A=\frac{3}{25},\ B=\frac{4}{25}$

$I=\frac{25}{25}\int e^{x/2}\left(3(3\cos x+4\sin x)^{2/3}+\frac{4\cdot(-3\sin x+4\cos x)}{3(\cos x+4\sin x)^{1/3}}\right)dx$

$6(3\cos x+4\sin x)^{2/3}e^{x/2}-4\int\frac{(-3\sin x+4\cos x)}{(3\cos x+4\sin x)^{1/3}}\cdot e^{x/2}dx+4\int\frac{(-3\sin x+4\cos x)}{(3\cos x+4\sin x)^{1/3}}\cdot e^{x/2}dx+C$

$\Rightarrow f(x)=6(3\cos x+4\sin x)^{2/3}e^{x/2}$

3. Put $x=2\sin\theta$

$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{\ln(2-2\sin\theta)}{2\cos\theta}\cdot 2\cos\theta d\theta\ I=\ln 2\cdot\pi+\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\ln(1-\sin\theta)d\theta\Rightarrow I=\pi\ell n2+I'$ Let

$I'=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\ln(1-\sin\theta)d\theta$

$\Rightarrow 2I'=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\ln(\cos^2\theta)d\theta=4\int_0^{\frac{\pi}{2}}\ln(\cos\theta)d\theta\Rightarrow I'=-\pi\ln 2$

4. $2z-\frac{2i}{z}=3\ 4z^2-\frac{4}{z^2}-8i=9\ 4z^2-\frac{4}{z^2}=9+8i\ 16\left(z^4+\frac{1}{z^4}\right)=49+144i$

$\Rightarrow z^8+1=\left(\frac{49+144i}{16}\right)z^4\Rightarrow z^{19}+z^{11}=\left(\frac{49+144i}{16}\right)\cdot z^{15}\Rightarrow\left(\frac{\alpha^{19}+\beta^{19}+\alpha^{11}+\beta^{11}}{\alpha^{15}+\beta^{15}}\right)=\frac{49-144i}{16}$

5. Let $P(m),P(p),P(c)$ be the probability of selecting book of maths, physics and chemistry respectively.

Clearly $P(m)=P(p)=P(c)=\frac{1}{3}$

Again let $P(s_1)$ and $P(s_2)$ be the probability that he solves the 1st and 1st as well as 2nd problem respectively.

Then

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$$P(s_1)=P(m)\times P\left(\frac{s_1}{m}\right)+P(p)\times P\left(\frac{s_1}{p}\right)+P(c)\times P\left(\frac{s_1}{c}\right)$$

$$\Rightarrow P(s_1)=\frac{1}{3}\times\frac{1}{2}+\frac{1}{3}\times\frac{3}{5}+\frac{1}{3}\times\frac{4}{5}=\frac{19}{30}$$

6. Similarly

$$P(s_2)=\frac{1}{3}\times\left(\frac{1}{2}\right)^2+\frac{1}{3}\times\left(\frac{3}{5}\right)^2+\frac{1}{3}\times\left(\frac{4}{5}\right)^2=\frac{125}{300}$$

$$\Rightarrow P\left(\frac{s_2}{s_1}\right)=\frac{\frac{125}{300}}{\frac{19}{30}}=\frac{25}{38}$$

7. $\left(\sum a_i\right)^2=\sum a_i^2-2\sum a_i a_j$

$\Rightarrow \sum a_i^2\le 3$

But by AM/GM

$$\Rightarrow \frac{\sum a_i^2}{n}\ge\left(\prod a_i^2\right)^{\frac{1}{n}}$$

$\sum a_i^2\ge n\Rightarrow n\le 3$

Possible polynomials are

$x+1, x-1, x^2+x-1, x^2-x-1$

x^3-x^2-x+1, x^3+x^2-x-1

8. Correct option answer is 3.00

9. Let $P(a\sec\theta, b\tan\theta)$ be the centre of the circle.

Let $(x_1,y_1),(x_2,y_2),(x_3,y_3)$ and (x_4,y_4) are the points of intersection of circle and the rectangular hyperbola $xy=c^2$.

So $\frac{x_1+x_2+x_3+x_4}{4}=\frac{a}{2}\sec\theta$

and $\frac{y_1+y_2+y_3+y_4}{4}=\frac{b}{2}\tan\theta$ [Using property] Let (h,k) be the centroid of triangle formed by ordered pairs $(x_1,y_1),(x_2,y_2),(x_3,y_3)$

$\Rightarrow 3h=x_1+x_2+x_3=2a\sec\theta-x_4$

$3k=y_1+y_2+y_3=2b\tan\theta-y_4\Rightarrow x_4=2a\sec\theta-3h$

$y_4=2b\tan\theta-3k$

Since (x_4,y_4) also lies on the $xy=c^2$.

$\Rightarrow (2a\sec\theta-3h)(2b\tan\theta-3K)=c^2$.

$$\Rightarrow\left(h-\frac{2a}{3}\sec\theta\right)\left(k-\frac{2b}{3}\tan\theta\right)=\left(\frac{c}{3}\right)^2$$

Hence locus is $\left(x-\frac{2a}{3}\sec\theta\right)\left(y-\frac{2b}{3}\tan\theta\right)=\left(\frac{c}{3}\right)^2$

Which is a rectangular hyperbola with centre $\left(\frac{2a}{3}\sec\theta,\frac{2b}{3}\tan\theta\right)$ which lies on the hyperbola

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=\left(\frac{2}{3}\right)^2$$

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10. Let $P(a\sec\theta, b\tan\theta)$ be the centre of the circle.

Let $(x_1,y_1),(x_2,y_2),(x_3,y_3)$ and (x_4,y_4) are the points of intersection of circle and the rectangular hyperbola $xy=c^2$.

So $\frac{x_1+x_2+x_3+x_4}{4}=\frac{a}{2}\sec\theta$ and $\frac{y_1+y_2+y_3+y_4}{4}=\frac{b}{2}\tan\theta$ [Using property] Let (h,k) be the centroid of triangle formed by ordered pairs $(x_1,y_1),(x_2,y_2),(x_3,y_3)$

$\Rightarrow 3h=x_1+x_2+x_3=2a\sec\theta-x_4$

$3k=y_1+y_2+y_3=2b\tan\theta-y_4$

$\Rightarrow x_4=2a\sec\theta-3h$

$y_4=2b\tan\theta-3k$

Since (x_4,y_4) also lies on the $xy=c^2$.

$\Rightarrow (2a\sec\theta-3h)(2b\tan\theta-3K)=c^2.\Rightarrow\left(h-\frac{2a}{3}\sec\theta\right)\left(k-\frac{2b}{3}\tan\theta\right)=\left(\frac{c}{3}\right)^2$

Hence locus is $\left(x-\frac{2a}{3}\sec\theta\right)\left(y-\frac{2b}{3}\tan\theta\right)=\left(\frac{c}{3}\right)^2$

Which is a rectangular hyperbola with centre

$\left(\frac{2a}{3}\sec\theta,\frac{2b}{3}\tan\theta\right)$ which lies on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=\left(\frac{2}{3}\right)^2$

11. Let O is the origin G is centroid of tetrahedron.

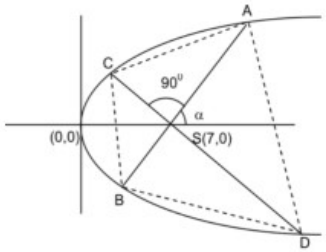
$(AB)^2+(AC)^2+(AD)^2+(BC)^2+(BD)^2+(CD)^2=16-16(OG)^2$

$\Rightarrow 16-16(OG)^2\geq 6(AB\bullet AC\bullet AD\bullet BC\bullet BD\cdot CD)^{1/3}\Rightarrow 16-16(OG)^2\geq 6\cdot\frac{2^3}{3}=16\Rightarrow OG=O\Rightarrow O$ and G are same points. So ABCD is a regular tetrahedron.

12. Since $y^2=28x$ is the parabola, than equation of a focal chord AB which is inclined at an angle α with x -axis is

$y-0=\tan\alpha(x-7)$

This satisfies $y^2=28x$



$\Rightarrow (x-7)^2\tan^2\alpha-28x=0$

$\Rightarrow x^2\tan^2\alpha-14x(\tan^2\alpha+2)+49\tan^2\alpha=0\#(2)$

If $A=(x_1,y_1)$ and $B=(x_2,y_2)$, then x_1,x_2 are the roots of equation (2)

Now $x_1+x_2=\frac{14(\tan^2\alpha+2)}{\tan^2\alpha}$ and $x_1x_2=49$

$x_1-x_2=\sqrt{(x_1+x_2)^2-4x_1x_2}$

$=\sqrt{\frac{14^2(\tan^2\alpha+2)^2}{\tan^4\alpha}-49\times 4}$

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$x_1x_2 = 28\cot\alpha\operatorname{cosec}\alpha$ and $y_1y_2 = 28\operatorname{cosec}\alpha$

$AB = 28\operatorname{cosec}^2\alpha$ Similarly, $CD = 28\sec^2\alpha$

Now area of $ACBD = \frac{1}{2}AB\,.\,CD$

$= \frac{1}{2}(28)^2\operatorname{cosec}\alpha\cdot\sec^2\alpha = \frac{784}{2}\operatorname{cosec}^2\alpha\sec^2\alpha = 1568\operatorname{cosec}^2 2\alpha$

13. $(2ny + xy\log x)\,dx = x\log x\,dy \Rightarrow \frac{dy}{y} = \left(\frac{2n}{x\log x} + 1\right)dx \Rightarrow \log(y) = 2n\log|\log x| + x + c$ (Since the curve passes through (e, e^e) we get $c = 0$)

$\Rightarrow y = e^{x+\log(\log x)^{2n}} \Rightarrow y = e^x\cdot e^{\log(\log x)^{2n}} \Rightarrow y = e^x(\log x)^{2n} \Rightarrow f(x) = e^x(\log x)^{2n}$

$$\Rightarrow g(x) = \lim_{n\to\infty} f(x) = \begin{cases} h \to \infty, & x \leq \frac{1}{e} \quad \because (\log x)^{2n} \to \text{infinity} \\ 0, & \frac{1}{e} < x < e \quad (\log x)^{2n} \to 0 \\ k \to \infty, & x \geq e \quad (\log x)^{2n} \to \text{infinity} \end{cases} \Rightarrow \int_{1/e}^{e} g(x)\,dx = 0$$

14. Trace of $AB - BA$ is O so let $AB - BA = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$

$$\Rightarrow (AB-BA)^2 = \begin{bmatrix} a^2+bc & 0 \\ 0 & a^2+bc \end{bmatrix} = (a^2+bc)I = KI$$

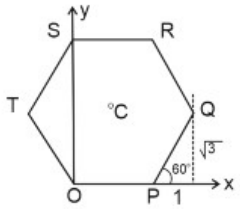
$\Rightarrow (AB-BA)^3 = K(AB-BA) \neq I$

So n cannot be odd number $\because$ trace of every odd power of $AB - BA$ is O

So let $n = 2m$

$\Rightarrow (AB-BA)^n = (K)^m I \Rightarrow (K)^m = 1 \Rightarrow K = 1$ or -1

15.



$O(0,0,0)$, $P(2,0,0)$, $Q(3,\sqrt{3},0)$, $R(2,2\sqrt{3},0)$,

$S(0,2\sqrt{3},0), T(-1,\sqrt{3},0)\ C(1,\sqrt{3},0) \Rightarrow U(1,\sqrt{3},\sqrt{5})\ \overrightarrow{OP} = 2\hat{i},\ \overrightarrow{OU} = \hat{i}+\sqrt{3}\,\hat{j}+\sqrt{5}\,\hat{k}$

$\cos\theta = \frac{|2|}{2\cdot\sqrt{1+3+5}} = \frac{1}{3}$ Equation of plane OUS is $\begin{vmatrix} x & y & z \\ 1 & \sqrt{3} & \sqrt{5} \\ 0 & 2\sqrt{3} & 0 \end{vmatrix} = 0 \Rightarrow \sqrt{5}x - z = 0$

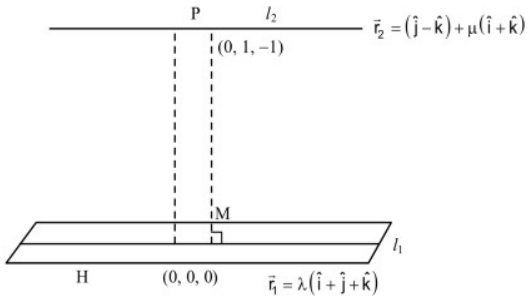
Equation of line $OU: \vec{r} = \lambda\left(\hat{i}+\sqrt{3}\,\hat{j}+\sqrt{5}\,\hat{k}\right)$

Equation of line $PQ: \vec{r} = 2\hat{i} + \mu\left(\hat{i}+\sqrt{3}\hat{j}\right)$

$\vec{b}_1\times\vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \sqrt{3} & \sqrt{5} \\ 1 & \sqrt{3} & 0 \end{vmatrix} = -\sqrt{15}\,\hat{i}+\sqrt{5}\,\hat{j}\ \vec{a}_2 - \vec{a}_1 = 2\hat{i}$ So shortest distance $= \frac{|-2\cdot\sqrt{15}|}{\sqrt{15+5}} = \sqrt{3}$

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16.



Line ℓ_2 is parallel to plane containing ℓ_1

Let ℓ, m, n be D. cosine of plane H

$l + m + n = 0 \#(1)$

$l + n = 0 \#(2)$

$-n + m + n = 0,\ m = 0\ \frac{\ell}{\sqrt{2}} = \frac{n}{-\sqrt{2}} = \frac{m}{0}\ \frac{\ell}{\sqrt{2}} = \frac{m}{0} = \frac{n}{-\sqrt{2}}$ So equation of plane H_0 is

$\sqrt{2}(x-0) + 0(y-0) - \sqrt{2}(z-0) = 0\ x - z = 0$ equation of plane H.

(A) $d[H_0] = PM = \frac{0+1}{\sqrt{1+1}} = \frac{1}{\sqrt{2}}$

$(B) = \left|\frac{0-2}{\sqrt{2}}\right| = \sqrt{2}$ (C) Distance of origin from $H_0 = 0$

(D) Distance of origin from the point of intersection of planes $y = z, x = 1, x = z$.

Intersection $T(1,1,1)$, Distance from origin $= \sqrt{1+1+1} = \sqrt{3}$

17. The derivative of $\ln(x^2 + 4)$ satisfies

$f'(x) = \frac{2x}{x^2+4} \le \frac{2|x|}{x^2+4} = \frac{\sqrt{4x^2}}{x^2+4} \le \frac{1}{2}$,

by the inequality between geometric and arithmetic mean. The important point here is that we have a strict inequality $f'(x) < 1/2$ for all $x \in R$ except for the single point $x = 2$. That suffices to conclude that

$f(b) - f(a) = \int_a^b f'(x)dx < \int_a^b \frac{1}{2}dx = \frac{1}{2}(b-a)$

for $a < b$, and therefore

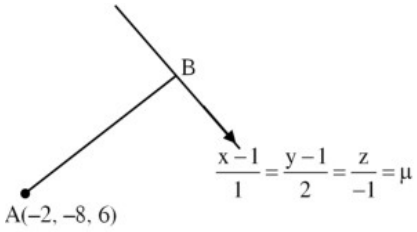
$$\left[8\frac{f(b)-f(a)}{b-a}\right] \le 3$$

18. $f'(x) = 3x^2 + 2bx + 3 + \cos x \forall x \in R \Rightarrow 3x^2 + 2bc + 2 \ge 0 \forall x \in R$

$\Rightarrow 4b^2 - 4.3.2 \le 0$

$b^2 \le 6$

19.



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DR or AB is $1,-1,2$

$\therefore$ Line AB is $\frac{x+2}{1}=\frac{y+8}{-1}=\frac{z-6}{2}=\lambda$

B is $(\lambda-2,-\lambda-8,2\lambda+6)$

B is $(\mu+1,2\mu+1,-\mu)$

Solving point B is $(-3,-7,4)$

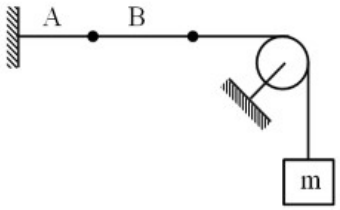
$AB=\sqrt{(-1)^2+(1)^2+(2)^2}=\sqrt{6}$

$AB^2=6$

PHYSICS

20. More magnetic force in axial as compared to equatorial arrangement. In alternate axial arrangement, force is repulsive $\Rightarrow$ more PE.

21.



$T=mg=9\ N$

$v_A=\sqrt{\frac{T}{\mu_A}}=\sqrt{\frac{9}{10^{-2}}}=30\ m/s$

$v_B=\sqrt{\frac{T}{\mu_B}}=\sqrt{\frac{9}{25\times10^{-3}}}=60\ m/s$

$f=\frac{nv_A}{2\ell_A}=\frac{kv_B}{2\ell_B}\Rightarrow\frac{n}{0.3}=\frac{2k}{0.5}\Rightarrow 5n=6k$

$\Rightarrow n=6,k=5,f_{min}=\frac{6\times30}{2\times0.3}=300\ Hz$

22. Correct Answer is (A)

23. $\frac{P}{4\pi r^2}\cdot\pi R^2\times\frac{1}{C}=\frac{GM_sm}{r^2}$

$\Rightarrow r\propto R^0$

24. $V_x=\frac{RT}{eN_A}m\left(\frac{150}{5}\right)$

$=\frac{\frac{25}{3}\times310\ln(30)}{1.6\times10^{-19}\times6\times10^{23}}=\frac{25\times31}{1.6\times18}\times3.4mV=91.5mV$

$e(V_x-V_0)=\frac{1}{2}\times39\times1.66\times10^{-27}\times V^2$

$1.6\times10^{-19}\times21.5\times10^{-3}=\frac{1}{2}\times39\times1.66\times10^{-27}\times V^2$

$V^2=\frac{1.6\times21.5\times2}{39\times1.66}\times10^5=10.63\times10^4$

$V=326\ m/s$

25. $V_x=\frac{RT}{eN_A}m\left(\frac{150}{5}\right)=\frac{\frac{25}{3}\times310\ln(30)}{1.6\times10^{-19}\times6\times10^{23}}$

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$$=\frac{25\times31}{1.6\times18}\times3.4\text{mV}=91.5\text{mV}$$

$$e\left(V_x-V_0\right)=\frac{1}{2}\times39\times1.66\times10^{-27}\times V^2$$

$$1.6\times10^{-19}\times21.5\times10^{-3}=\frac{1}{2}\times39\times1.66\times10^{-27}\times V^2$$

$$V^2=\frac{1.6\times21.5\times2}{39\times1.66}\times10^5=10.63\times10^4$$

$$V=326\text{ m/s}$$

$$V_{avg}=\frac{V}{2}$$

Distance covered $=\frac{D_A}{2}$

Time taken, $t=\frac{D_A/2}{V/2}=\frac{D_A}{V}=30.67\text{ ns}$

26. $$\frac{T-T_0}{\left(\frac{d}{\lambda.6ab}+\frac{d}{\lambda.18ab}\right)}=P_0$$

$$T=T_0+P_0\cdot\frac{2\,d}{9\lambda ab}$$

27. $$\frac{T-T_0}{\frac{d}{6\lambda ab}+\frac{d}{18\lambda ab}+\frac{d}{30\lambda ab}}=P_0$$

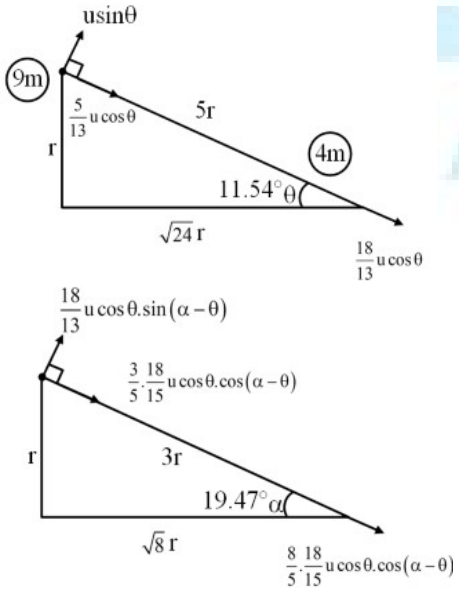
$$T=T_0+\frac{dP_0.23}{90\lambda ab}$$

28. $$\lambda_{PP}=\frac{340}{200\times\frac{340}{325}}=1.625$$

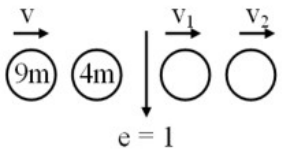
29. $$\lambda_P=200\times\frac{340+30}{340-15}=200\times\frac{370}{325}=227.69$$

$$\frac{310\times13}{8\times370}=1.36\text{ m}$$

30.



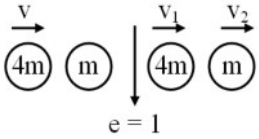
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$9mv_1 + 4mv_2 = 9mv$

$v_1 + \frac{4}{9}v_2 = v; v_2 - v_1 = v$

$v_2 = \frac{18}{13}v, v_1 = \frac{5}{13}v$



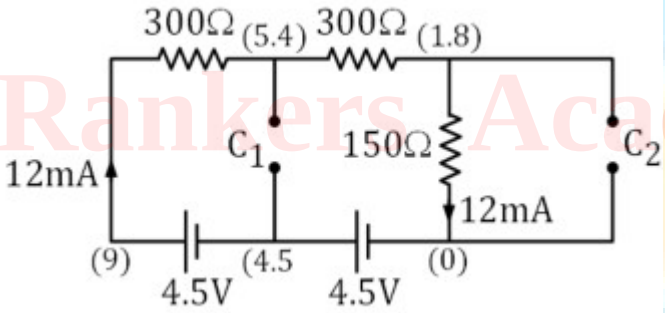
$4mv_1 + mv_2 = 4mv$

$\Rightarrow v_1 + \frac{v_2}{4} = v\#(1)$

$v_2 - v_1 = v\#(2)$

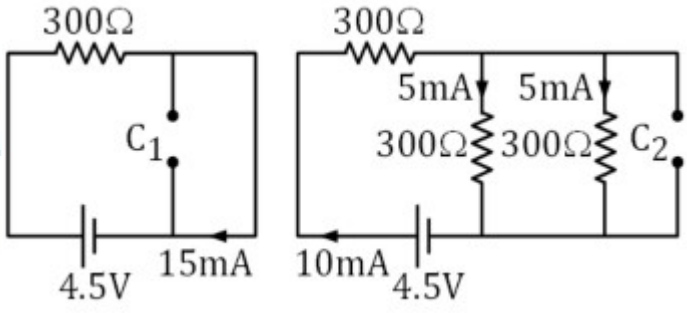
$v_2 = \frac{8}{5}v, v_1 = \frac{3}{5}v$

31. (A) When S_1 is open and S_2 is closed for long time



Charge on $C_1 = 4\mu Fa(5.4 - 4.5)v = 3.6\mu C$

(C) & (D) we close switch S_1 at $t = 0$, and after long time :



$\Rightarrow$ Current through $S_1 = 5$ mA and charge on

$C_2 = 4\mu$ F $\times 1.5$ V $= 6\mu C$

32. $Q_1 = (232.0381 - 228.031) - 4.0026) \times 931MeV$

$= (0.0381 - 0.0337) \times 931MeV$

$= 0.0044 \times 931MeV = 4.10MeV$

$\Rightarrow KE_\alpha = 4.10 - 0.12 - 0.18MeV = 3.8MeV$

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$Q_2 = (228.0311 - 228.0290) \times 931 MeV$

$= 0.0021 \times 931 MeV = 1.96 MeV$

$E_{\beta_{max}} = Q_2 - -0.08 - -0.20 - -0.35 MeV$

$= 1.96 - 0.63 MeV = 1.33 MeV$

33. Rules of significant figures

34. Screw gauge : $LC = \frac{0.5mw}{50} = 0.01$ mm

$OD = 4.23 + 0.002 = 4.25$ mm

$ID = 2.50 + 0.45 + 0.02 = 2.97$ mm

$VC : LC = 1 - \frac{9}{10}$ mm $= 0.1$ mm

$L = 102 + 6 \times 0.1 - 0.1 = 102.5$ mm

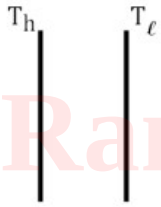
$$\text{density} = \frac{\text{mass}}{\text{volume}} = \frac{m}{\frac{\pi L}{4}(OD^2 - ID^2)}$$

$$\frac{\Delta\delta}{\delta} = \frac{\Delta V}{V} = \frac{\Delta L}{L} + \frac{\Delta(OD^2 - ID^2)}{OD^2 - ID^2}$$

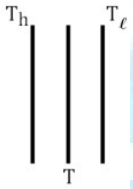
$$= \frac{0.1}{102.5} + \frac{2 \times 4.25 \times 0.01 + 2 \times 2.97 \times 0.01}{4.25^2 - 2.97^2}$$

$= 0.10\% + 1.5625\% = 1.66\%$

36.



$H_1 = \sigma A(T_h^4 - T_\ell^4)$



$H_2 = \sigma A(T_h^4 - T^4) = \sigma A(T^4 - T_\ell^4)$

$\Rightarrow 2T^4 = T_h^4 + T_\ell^4$

$$\Rightarrow H_2 = \sigma A\left(T_h^4 - \frac{T_h^4 + T_\ell^4}{2}\right)$$

$$= \frac{\sigma A}{2}(T_h^4 - T_\ell^4) \Rightarrow \frac{H_1}{H_2} = 2$$

37. $\frac{1}{f_\ell} = \left(\frac{1.5-1}{1}\right)\left(\frac{1}{20} - \frac{1}{60}\right) = \frac{1}{60}$

$\frac{1}{60} = \frac{1}{v_1} - \frac{1}{-80} \Rightarrow v_1 = 240$ cm

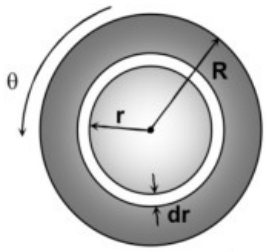
Due to slab image shifts to right by 1 cm

$\Rightarrow 8 - 1 + 3 + x = 20$ ($\because$ image forms at C.O.C. of mirror after two refractive)

$\Rightarrow x = 10$

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38.



$$dU = \frac{1}{2}\eta\left(\frac{r\theta}{\ell}\right)^2 2\pi r dr\ell$$

$$\int dU = \frac{r\eta\theta^2}{\ell}\int_0^R r^3 dr$$

$$U = \frac{\pi\eta R^4\theta^2}{4\ell}$$

The elastic deformation energy stored in the rod is

$$U = \frac{\pi\eta R^4\theta^2}{4\ell}$$

CHEMISTRY

40. (B) Radial node will be spherical

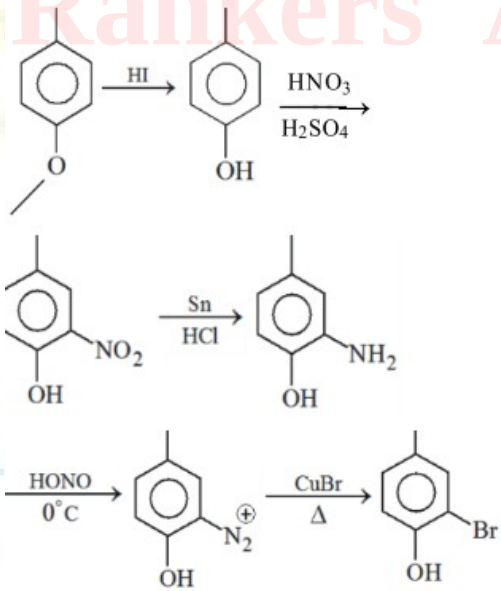
41. (A) Only Al^{+3} produces solution with NaOH excess

(B) Red $\rightarrow \left[Fe(H_2O)_5(SCN)\right]^{2+}$ but reaction is no redox

(C) BiOCl ↓ soluble in dilute HCl

(D) SnS is soluble in YAS

42.



43. The correct answer is 27.93

44. The correct answer is 514.00

45. $x_1 = 7$ and $x_2 = 2$

(Except 2,3) (only 1,10)

46. $x_3 = 7$ and $x_4 = 7$

(Except 2,3)

(Except 1,2)

50. (A) Fact

(B) Sudden rise in heat capacity due to formation of defect.

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(C) Decrease in density.
(D) Mainly due to increase in configurational entropy.

51. The correct answer is option (A), (C), (D)

52. (A) Head to tail favoured for nearly spherical molecules.
(C) Not all are dependent on temperature.

53. The correct answer is option (A, B, C)

54. Explanation Question:
Asking about the correct orders for the given species
Concept:
This question Based on application of electronic displacement efects
Solution:
(1) Acdic strength $\left(K_a\right) \propto$ Stability of conjugate base $\propto \frac{-m,-I}{+m,+I}$
(2) Stability of carbocations
$\propto \frac{-M,+H,+I}{-M,-I,\ \text{Bridged carbocation}}$.
(3) Stability of carbene; Triplet $\underset{(1CH_2)}{1}{}^{1} >$ singlet $\underset{(CH_2)}{11}$
(4) Basic strength $\left(K_b\right)$: (in polar solvent) : 2° amine $> 1^{\circ}$ amine 3° amine $> NH_3$; (1 alkyl group is methyl) According to above concepts except option : CH_2 (singlet) $>: CH_2$ (triplet) (stability order) all other options are correct.
Final Answer: The correct option is (1), (2) and (4).

55. $\Delta G = -nFE^{\circ}_{cell}$
$n = \left(4 - \left(-\frac{9}{4}\right)\right) \times 8 = 50$
$E^{\circ}_{cell} = \frac{3860 \times 1000}{50 \times 96500}$
$= .80\ V = 800mV$

56. $XeO_3 + XeOF_4 \rightarrow XeO_2\ F_2$
Total number of d-orbital use in bonding in product
(A) is 3
Total number of $p\pi - p\pi$ bond present in product
(A) is 0
$(3+0) = 3$

57. Question Explanation: The question asks to calculate the percentage of the α-anomer present in an equilibrium mixture using optical rotation values.
Concept : This question is based on Mutarotation, Optical activity, Anomeric equilibrium
Solution:
Let the amount of α-form in equilibrium mixture is X . then the amount of β-form in the equilibrium mixture is $(1-X)$.
so,
$150.7(X) + 52.8(1-X) = +80.2$
$150.7X + 52.88 - 52.8X = +80.2$
$97.9X = 27.4$
$X = 0.279 = 0.28$
$\%X = 28\%$

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